## Question

Consider the $H_2O$ molecule with $C_{2v}$ symmetry, where the potential energy surface of its highly excited vibrational states exhibits significant anharmonicity and vibration-rotation coupling. Considering higher-order perturbations (such as Darling-Dennison resonance, Fermi resonance, Coriolis coupling) and isotopic substitution, select the statement that does not contain a correct description.

1. Under this symmetry, the symmetric stretching $\nu_1$ and antisymmetric stretching $\nu_3$ both belong to the $A_1$ irreducible representation, so their Fermi resonance must satisfy $\Delta v = 0$.

2. The $v = 2$ energy level of the bending vibration $\nu_2$ can be coupled with $\nu_1 + \nu_3$ through generalized Darling-Dennison resonance, which needs to satisfy the rotational selection rule of $\Delta K_a = \pm 2$.

3. The substitution of the $\mathrm{H}$ atom with a $\mathrm{D}$ atom will significantly reduce the frequency of $\nu_3$, but the frequency change of $\nu_2$ is mainly determined by anharmonicity corrections.

4. In this molecule, there is no rotation-induced Coriolis coupling interaction between $\nu_1$ and $\nu_3$.

5. In the rovibrational spectrum of $H_2{}^{18}O$, the $Q$ branch of the $\nu_3$ band is almost completely forbidden due to Coriolis coupling, but the $Q$ branch appears in $HD^{18}O$ obtained after isotopic substitution due to the reduction in symmetry.

6. If $\nu_1$ and $2\nu_2$ undergo Fermi resonance, its resonance intensity is proportional to the cubic anharmonic coefficient $k_{122}$. In $D_2O$, the vibrational frequency of $\nu_2$ decreases, and the resonance disappears.

7. Higher-order centrifugal distortion terms (such as $H_{JK}$) lead to $K_a-$dependent splitting in the rotational energy levels of $H_2O$, and its sign may be opposite to that of $D_2O$ due to the different mass dependence of rotational constants $B$ and $C$.

8. In the $\nu_2$ excited state $v2 = 1$ of $H_2O$, the energy gap $\Delta E$ of $l-$type doubling $(l = \pm 1)$ can be estimated by second-order perturbation theory: $\Delta E \propto \zeta_2^2 (J+1)/B$, where $\zeta_2$ is the Coriolis coupling coefficient.

9. The hyperfine structure of $H_2{}^{17}O$ originates from nuclear spin-rotation coupling, and the $F-$branch splitting interval of its $\nu_3$ band is linearly related to $J$.

10. In the $\nu_1 + \nu_3$ combination band of $\mathrm{H_2O}$, the rotational transition intensity of $\Delta K_a = \pm 1$ is modulated by the Herman-Wallis factor, and its functional form is $(1 + aK_a^2)$.

A. The remaining options do not meet the criteria.

B. 1, 2, 3, 4, 5

C. 1, 3, 5, 8, 10

D. 2, 4, 5, 6, 7

E. 3, 6, 7, 8, 9

F. 1, 3, 4, 6, 9

G. 4, 7, 8 ,9 ,10

H. 2, 5, 7, 9, 10

## Answer

Correct Answer: **F**

## Detailed Explanation

The $H_2O$ molecule with $C_{2v}$ symmetry has three vibrational modes: the symmetric stretching vibration $\nu_1$ and the bending vibration $\nu_2$ both belong to the $A_1$ irreducible representation, and the antisymmetric stretching vibration $\nu_3$ belongs to the $B_2$ irreducible representation. Furthermore, $\Delta\nu = 0$ is not a condition for Fermi resonance. Statement 1 is incorrect.

**CHECKPOINT** 0.5 PTS

Vibration $\nu_3$ belongs to the $B_2$ irreducible representation, and $\Delta\nu = 0$ is not a condition for Fermi resonance. Statement 1 is incorrect.

Under $C_{2v}$ symmetry, $2\nu_1$ belongs to $A_1 \otimes A_1 = A_1$, and $\nu_1 + \nu_3$ belongs to $A_1 \otimes B_2 = B_2$. The symmetries of the two do not match. For the $H_2O$ molecule, $K_a$ is the rotational quantum number around the $C_2$ axis. When it is even, the symmetry of the rotational wave function is $A_1$ or $A_2$, and the eigenvalue for the $C_2$ operation is $+1$. When it is odd, the symmetry of the rotational wave function is $B_1$ or $B_2$, and the eigenvalue for the $C_2$ operation is $-1$. When a rotational transition of $\Delta K_a = \pm 2$ occurs, the same parity of $\Gamma_{rotation}$ is maintained. At this time, the symmetry of the vibrational state becomes $A_1 \otimes \Gamma_{rotation}$ and $B_2 \otimes \Gamma'_{rotation}$, introducing a non-zero vibration-rotation cross term, making the transition allowed. Statement 2 is correct.

**CHECKPOINT** 0.5 PTS

Rotational transitions of $\Delta K_a = \pm 2$ introduce a non-zero vibration-rotation cross term, making the transition allowed. Statement 2 is correct.

The basic formula for vibrational frequency is $\nu = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$. The substitution of the $\mathrm{D}$ atom for the $\mathrm{H}$ atom increases the reduced mass $\mu = \frac{m_{\mathrm{O}} m_{\mathrm{H}}}{m_{\mathrm{O}} + m_{\mathrm{H}}}$ by approximately 89%, and the $\nu_3$ frequency decreases by approximately 27%. For the bending vibration $\nu_2$, it is obtained through constructing a geometric model that $\mu \approx \frac{m_{\mathrm{H}} m_{\mathrm{O}}}{2(M_{\mathrm{O}} + 2M_{\mathrm{H}})} \times r^2 \times \cos^2\frac{\theta}{2}$. Therefore, the substitution of the $D$ atom for the $H$ atom increases the reduced mass by approximately 81%, and the $\nu_2$ frequency decreases by approximately 26%. From this, it can be seen that the frequency changes of $\nu_2$ and $\nu_3$ are mainly caused by changes in the reduced mass, rather than nonlinear corrections. Statement 3 is incorrect.

**CHECKPOINT** 0.5 PTS

The frequency changes of $\nu_2$ and $\nu_3$ are mainly caused by changes in the reduced mass, rather than nonlinear corrections. Statement 3 is incorrect.

Coriolis coupling requires that the direct product of the two vibrational modes contains the symmetry of the angular momentum component, i.e., $\Gamma(\hat{J}_a) \subset \Gamma(\psi_i) \otimes \Gamma(\psi_j)$. For the two vibrational states $\nu_1$ and $\nu_3$, we have $A_1 \otimes B_2 = B_2$, and the rotation operator $\hat{J}_Y$ belongs to $B_2$ symmetry, so $\nu_1$ and $\nu_3$ can be coupled through $\hat{J}_Y$. Statement 4 is incorrect.

**CHECKPOINT** 0.5 PTS

$\nu_1$ and $\nu_3$ can be coupled through $\hat{J}_Y$. Statement 4 is incorrect.

The antisymmetric stretching vibration $\nu_3$ has intrinsic angular momentum, and Coriolis coupling causes an increase in vibrational angular momentum, so the rotational angular momentum must decrease accordingly. However, the $Q$ branch requires that the rotational angular momentum remains unchanged, i.e., $\Delta J = 0$, so it is almost completely forbidden. The symmetry of $\mathrm{HD^{18}O}$ obtained after isotope substitution is reduced from $C_{2v}$ to $C_s$, and all vibrational modes become $A'$. The symmetry constraint is greatly reduced, so the $Q$ branch can be observed. Statement 5 is correct.

**CHECKPOINT** 0.5 PTS

The $Q$ branch is almost completely forbidden because it requires the rotational angular momentum to remain unchanged; the symmetry of $HD^{18}O$ is reduced, allowing it. Statement 5 is correct.

The anharmonic Hamiltonian is $\hat{H}_{非谐} = \sum k_{i,j,k} \times q_i q_j q_k$, so the coupling matrix element of $\nu_1$ and $2\nu_2$ is $W = \langle \nu_1 | \hat{H}_{非谐} | \nu_2 \rangle = \langle \nu_1 | K_{122} q_1 q_2^2 | \nu_2 \rangle$, which is proportional to $k_{122}$. In $D_2O$, the isotope mass effect causes the frequencies of $\nu_1$ and $\nu_2$ to decrease almost proportionally, and the resonance is weakened due to the increase in the energy difference, but it still exists. Statement 6 is incorrect.

**CHECKPOINT** 0.5 PTS

Coupling matrix element $W = \langle \nu_1 | K_{122} q_1 q_2^2 | \nu_2 \rangle \propto k_{122}$. In $D_2O$, the resonance is weakened due to the increase in the energy difference, but it still exists. Statement 6 is incorrect.

The rotational Hamiltonian is in the form of $\hat{H}_{rot} = A\hat{J}_a^2 + B\hat{J}_b^2 + C\hat{J}_c^2 - D_J\hat{J}^4 - D_{JK}\hat{J}^2\hat{J}_a^2 - D_K\hat{J}_a^4 + H_{JK}\hat{J}^2\hat{J}_a^2 + \ldots$, where for the $H_{JK}$ term we have $H_{JK}\hat{J}^4 K_a^2$. Similarly, processing other terms, the relationship between the rotational energy level and $K_a$ can be expressed as $E(J, K_a) = BJ(J+1) + (C-B)K_a^2 - D_J J^2(J+1)^2 - D_K K_a^4 + H_{JK} K_a^2 J^4 + \ldots$, where higher-order centrifugal distortion terms such as $D_J, D_K, H_{JK}$ all introduce $K_a-$ dependent energy level splitting. The dependence of the rotational constant on mass is expressed through the moment of inertia, i.e., $B = \frac{\hbar^2}{2I_b}, C = \frac{\hbar^2}{2I_c}$. Considering the sixth-order dominant term $H_{JK} \propto \frac{(B-C)^3}{\omega^2}$, which depends on the rotational constants $B, C$, the mass effect brought about by isotope substitution causes a change in the magnitude of the moment of inertia, so $B - C$ may change sign, and the corresponding $H_{JK}$ changes sign accordingly. Statement 7 is correct.

CHECKPOINT 0.5 PTS

Sixth-order dominant term $H_{JK} \propto (B-C)^3$, the mass effect causes a change in the magnitude of the moment of inertia, $B-C$ may change sign, and $H_{JK}$ changes sign accordingly. Statement 7 is correct.

In the excited state of the bending vibration $\nu_2$, the $l-$type doubling $(l=\pm1)$ originates from Coriolis coupling. Rotational energy level mixing in the second-order perturbation theory introduces the intermediate state $|v_2=0,J'\rangle$, obtaining $\Delta E \approx 2\sum_{J'} \frac{\langle v_2=1,l=+1 \mid \hat{H}_{\text{Coriolis}}|v_2=0,J'\rangle\langle v_2=0,J' \mid \hat{H}_{\text{Coriolis}}|v_2=1,l=-1\rangle}{E_{v=1}-E_{v=0}}$. Where $\hat{H}_{\text{Coriolis}} = -2\sum_{\alpha=a,b,c} B_\alpha \zeta_\alpha^{(\Gamma)} J_\alpha l_\alpha$. For $\nu_2$ belonging to the $A_1$ irreducible representation, only $\zeta_b^{(A_1)} \propto \zeta_2 \neq 0$, obtaining $\hat{H}_{\text{Coriolis}} = -2B_b\zeta_2 J_b l_b$. Substituting $E_{v=1}-E_{v=0} \approx \hbar\omega_2, J'=\pm1$ and the quantization condition, we get $\Delta E \propto \frac{(2B_b\zeta_2 J_b*)^2}{\hbar\omega_2} \propto \frac{\zeta_2^2(J+1)}{B}$. Statement 8 is correct.

CHECKPOINT 0.5 PTS

Introducing the intermediate state $|v_2=0,J'\rangle$, substituting the energy gap and the quantization condition, we get $\Delta E \propto \frac{\zeta_2^2(J+1)}{B}$. Statement 8 is correct.

The nuclear spin of $^{17}O$ is $I=5/2$, which couples with the angular momentum $J$ of the molecular rotation to form a total angular momentum $F=J+I$, leading to hyperfine splitting of the rotational energy levels. The main term of the hyperfine Hamiltonian is $H_Q \propto 3(I\cdot J)^2 - I^2J^2$, expanding $(I\cdot J)^2 = \frac{1}{4}(F^2-I^2-J^2)$, substituting the energy level correction $\Delta E_Q \propto \frac{3}{4}K^2 - I(I+1)J(J+1)$, where $K=F(F+1)-I(I+1)-J(J+1)$, and $F$ is the total angular momentum quantum number. Therefore, the splitting spacing is not linearly related to $J$. Statement 9 is incorrect.

**CHECKPOINT** 0.5 PTS

Energy level correction $\Delta E_Q \propto \frac{3}{4}K^2 - I(I+1)J(J+1)$, therefore the splitting spacing is not linearly related to $J$. Statement 9 is incorrect.

The Herman-Wallis factor describes the change in rotational transition intensity with $K_a$. The sign and magnitude of the parameter $a$ are related to the rotational constant, vibration-rotation coupling, etc. of the molecule. For transitions of $\Delta K_a = \pm 1$, let the rotation-vibration wave function of the molecule be $|v, J, K_a\rangle$, the transition dipole moment element $M = \langle v', J', K_a'|\mu|v, J, K_a\rangle$. Considering the Herman-Wallis effect, the vibrational wave function can be expanded as $|v\rangle = |v_0\rangle + \alpha K_a^2|v_0\rangle$, so the transition dipole moment element is expanded as $M \approx \langle v_0', J', K_a'|\mu|v_0, J, K_a\rangle + \alpha K_a^2\langle v_0', J', K_a'|\mu|v_0, J, K_a\rangle$. The transition intensity $I \propto |M|^2 \approx I_0(1 + 2\alpha K_a^2)$. Therefore, the Herman-Wallis factor is $(1 + aK_a^2)$, where $a = 2\alpha$. Statement 10 is correct.

**CHECKPOINT** 0.5 PTS

Vibrational wave function expansion $|v\rangle = |v_0\rangle + \alpha K_a^2|v_0\rangle$, transition intensity $I \propto |M|^2 \approx I_0(1 + 2\alpha K_a^2)$. The Herman-Wallis factor is $(1 + aK_a^2)$, where $a = 2\alpha$. Statement 10 is correct.

The incorrect statements are 1, 3, 4, 6, 9, so choose F.

**CHECKPOINT** 0.5 PTS

The incorrect statements are 1, 3, 4, 6, 9, so choose F.